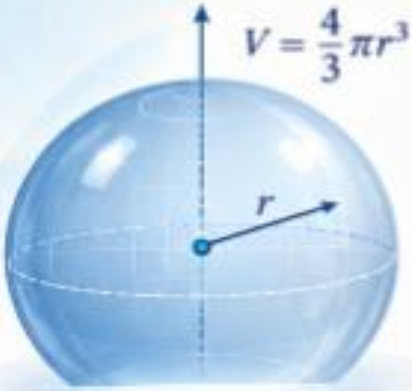
EBA0201 – Precalculus

Lecture 5

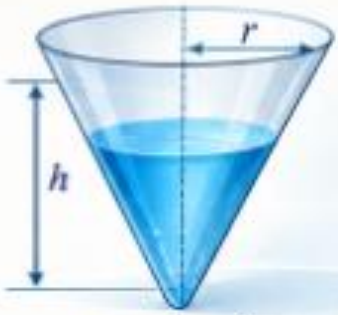
APPLICATIONS OF DERIVATIVES

TIME RATE

and Parametric Differentiation

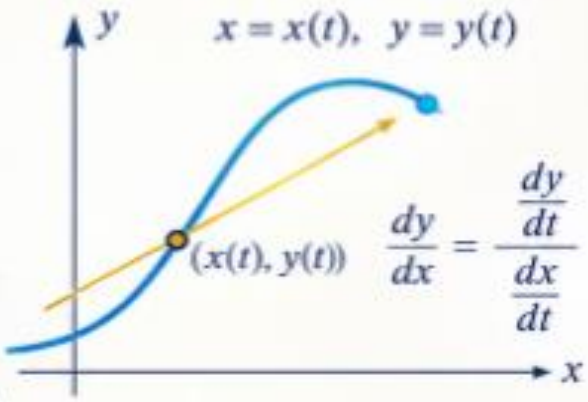


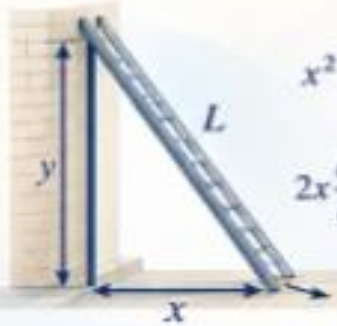
$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$



$$V = \frac{1}{3}\pi r^2 h$$

$$\frac{dV}{dt} = \frac{1}{3}\pi\left(2r\frac{dr}{dt}\,h + r^2\frac{dh}{dt}\right)$$





$$x^2 + y^2 = L^2$$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

- Area and Volume Rates
- Changing Dimensions
- Pythagorean Related Rates
- Similar Triangles and Cones
- Parametric Differentiation

Coordinator:

Dr. Taha Salah El-Din

Dr. Ola Hamada

MATHEMATICS
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1 Lesson Idea

When both x and y are written in terms of a parameter t, we can find the slope of the curve without eliminating t. We differentiate x and y with respect to t and use the chain rule.

2 Parametric Form

When a curve is given in parametric form, both coordinates are functions of the same parameter t:

$$x = x(t), \quad y = y(t)$$

3 First Derivative

The slope of the curve is given by

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

This formula is valid when $\frac{dx}{dt} \neq 0$.

4 How to Find the First Derivative

1. Differentiate x with respect to t to find $\frac{dx}{dt}$.
2. Differentiate y with respect to t to find $\frac{dy}{dt}$.
3. Divide $\frac{dy}{dt}$ by $\frac{dx}{dt}$.
4. Simplify the result.

5 Second Derivative

The second derivative is found using

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

6 How to Find the Second Derivative

1. First find $\frac{dy}{dt}$ using the method above.
2. Differentiate this result with respect to t, that is, find $\frac{d}{dt}\left(\frac{dy}{dx}\right)$.
3. Divide by $\frac{dx}{dt}$.
4. Simplify the result.

7 Allowed Functions in This Lecture

In this lecture, $x(t)$ and $y(t)$ may involve the following functions only:

- Polynomial functions
- Trigonometric functions
- Exponential functions involving e

Logarithmic functions are **not** included in this lecture.

8 Important Notes

1. Keep everything in terms of the parameter t. There is no need to eliminate t.
2. Simplify carefully, especially when dividing expressions.
3. Make sure $\frac{dx}{dt}$ is not zero, since the formula for $\frac{dy}{dx}$ (and for $\frac{d^2y}{dx^2}$) requires $\frac{dx}{dt} \neq 0$.

1 Example 1

Problem: Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if $x = t^2 + 1,\ y = t^3 - 3t.$

Solution:

$$\frac{dx}{dt} = \frac{d}{dt}(t^2 + 1) = 2t.$$

$$\frac{dy}{dt} = \frac{d}{dt}(t^3 - 3t) = 3t^2 - 3.$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2 - 3}{2t}.$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left[\frac{3t^2 - 3}{2t}\right] = \frac{d}{dt}\left[\frac{3}{2}\left(t - \frac{1}{t}\right)\right] = \frac{3}{2}\left(1 + \frac{1}{t^2}\right) = \frac{3(t^2 + 1)}{2t^2}.$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt} = \frac{\frac{3(t^2 + 1)}{2t^2}}{2t} = \frac{3(t^2 + 1)}{4t^3}.$$

First derivative:

$$\frac{dy}{dx} = \frac{3t^2 - 3}{2t}$$

Second derivative:

$$\frac{d^2y}{dx^2} = \frac{3(t^2 + 1)}{4t^3}$$

2 Example 2

Problem: Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if $x = \sin t,\ y = \cos^2 t.$

Solution:

$$\frac{dx}{dt} = \cos t.$$

$$\frac{dy}{dt} = \frac{d}{dt}(\cos^2 t) = 2\cos t(-\sin t) = -2\sin t\cos t.$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-2\sin t\cos t}{\cos t} = -2\sin t.$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}(-2\sin t) = -2\cos t.$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt} = \frac{-2\cos t}{\cos t} = -2.$$

First derivative:

$$\frac{dy}{dx} = -2\sin t$$

Second derivative:

$$\frac{d^2y}{dx^2} = -2$$

3 Example 3

Problem: Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if $x = e^t,\ y = te^t.$

Solution:

$$\frac{dx}{dt} = e^t.$$

$$\frac{dy}{dt} = \frac{d}{dt}(te^t) = e^t + te^t = e^t(1 + t).$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{e^t(1 + t)}{e^t} = 1 + t.$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}(1 + t) = 1. \quad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt} = \frac{1}{e^t} = e^{-t}.$$

First derivative:

$$\frac{dy}{dx} = 1 + t$$

Second derivative:

$$\frac{d^2y}{dx^2} = e^{-t}$$

4 Example 4

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if $x = t^2,\ y = e^t$.

Solution

$$\frac{dx}{dt} = 2t.$$

$$\frac{dy}{dt} = e^t.$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{e^t}{2t}.$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left[\frac{e^t}{2t}\right] = \frac{e^t(t-1)}{2t^2}.$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{e^t(t-1)}{2t^2}}{2t} = \frac{e^t(t-1)}{4t^3}.$$

$$\frac{dy}{dx} = \frac{e^t}{2t} \qquad \frac{d^2y}{dx^2} = \frac{e^t(t-1)}{4t^3}$$

5 Example 5

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if $x = \tan t,\ y = \sec t$.

Solution

$$\frac{dx}{dt} = \sec^2 t.$$

$$\frac{dy}{dt} = \sec t \tan t.$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sec t \tan t}{\sec^2 t} = \sin t.$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}(\sin t) = \cos t.$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\cos t}{\sec^2 t} = \cos^3 t.$$

$$\frac{dy}{dx} = \sin t \qquad \frac{d^2y}{dx^2} = \cos^3 t$$

6 Example 6

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ if $x = t^2,\ y = \sin t$.

Solution

$$\frac{dx}{dt} = 2t.$$

$$\frac{dy}{dt} = \cos t.$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{2t}.$$

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{d}{dt}\left[\frac{\cos t}{2t}\right] = -\frac{t\sin t + \cos t}{2t^2}.$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{-\frac{t\sin t + \cos t}{2t^2}}{2t} = -\frac{t\sin t + \cos t}{4t^3}.$$

$$\frac{dy}{dx} = \frac{\cos t}{2t} \qquad \frac{d^2y}{dx^2} = -\frac{t\sin t + \cos t}{4t^3}$$

1. Lesson Idea

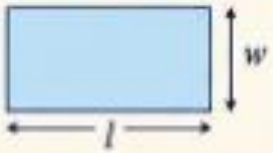
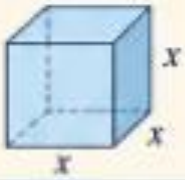
Area and volume rates describe how the area or volume of a geometric object changes with time when one or more dimensions are changing.

> Write the formula first, differentiate with respect to time t, then substitute the given values.

2. Four-Step Method

1. Define the changing variables.
2. Write one formula for area or volume.
3. Differentiate both sides with respect to t.
4. Substitute the known values and solve for the required rate.

3. Common Formulas

Shape	Formula	Time Derivative
Circle 	$A = \pi r^2$	$\frac{dA}{dt} = 2\pi r\left(\frac{dr}{dt}\right)$
Sphere 	$V = \frac{4}{3}\pi r^3$	$\frac{dV}{dt} = 4\pi r^2\left(\frac{dr}{dt}\right)$
Rectangle 	$A = lw$	$\frac{dA}{dt} = l\left(\frac{dw}{dt}\right) + w\left(\frac{dl}{dt}\right)$
Cube 	$V = x^3$	$\frac{dV}{dt} = 3x^2\left(\frac{dx}{dt}\right)$

4. Important Notes

1. Do not substitute numerical values before differentiating.
2. Keep the sign of each rate: increasing is positive, decreasing is negative.
3. Always include units in the final answer.

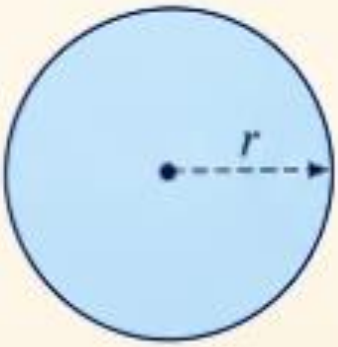
Next pages: Worked Examples 1–4 →

Worked Examples 1–2

1 Example 1 – Expanding Circle

? Problem

The radius of a circle increases at 3 cm/s. Find the rate of change of its area when $r = 5$ cm.



Solution

$A = \pi r^2$

Differentiate with respect to t:

$$\frac{dA}{dt} = 2\pi r \cdot \frac{dr}{dt}$$

Substitute $r = 5$ and $dr/dt = 3$:

$$\frac{dA}{dt} = 2\pi(5)(3)$$

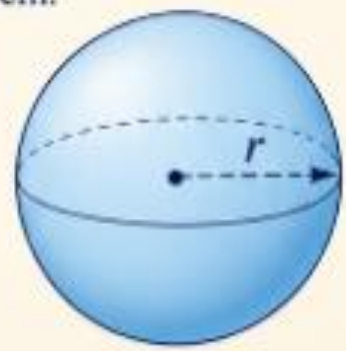
$$\frac{dA}{dt} = 30\pi \text{ cm}^2/\text{s}$$

Final Answer: $\dfrac{dA}{dt} = 30\pi \text{ cm}^2/\text{s}$

2 Example 2 – Expanding Sphere

? Problem

The radius of a sphere increases at 2 cm/s. Find dV/dt when $r = 4$ cm.



Solution

$V = \frac{4}{3}\pi r^3$

Differentiate with respect to t:

$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

Substitute $r = 4$ and $dr/dt = 2$:

$$\frac{dV}{dt} = 4\pi(4)^2(2)$$

$$\frac{dV}{dt} = 128\pi \text{ cm}^3/\text{s}$$

Final Answer: $\dfrac{dV}{dt} = 128\pi \text{ cm}^3/\text{s}$

Important Note

Units matter: area rates are in square units per second, volume rates are in cubic units per second.

Example 3 – Rectangle with Two Changing Sides

Problem

A rectangle has length $l = 10$ cm and width $w = 8$ cm. The length increases at 3 cm/s while the width decreases at 2 cm/s. Find dA/dt.



Solution

$A = lw$

Differentiate with respect to t:

$$\frac{dA}{dt} = l\frac{dw}{dt} + w\frac{dl}{dt}$$

Substitute $l = 10$, $w = 8$, $\frac{dl}{dt} = 3$, and $\frac{dw}{dt} = -2$:

$$\frac{dA}{dt} = 10(-2) + 8(3)$$

$$\frac{dA}{dt} = -20 + 24$$

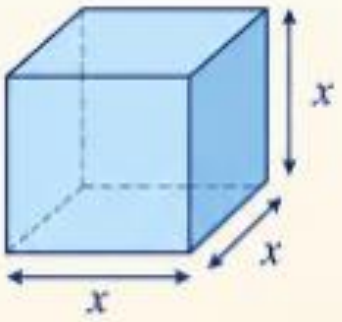
$$\frac{dA}{dt} = 4 \text{ cm}^2/\text{s}$$

Final Answer: $\frac{dA}{dt} = 4 \text{ cm}^2/\text{s}$

Example 4 – Expanding Cube

Problem

The edge x of a cube increases at 0.5 cm/s. Find the rate of change of volume when $x = 6$ cm.



Solution

$V = x^3$

Differentiate with respect to t:

$$\frac{dV}{dt} = 3x^2\frac{dx}{dt}$$

Substitute $x = 6$ and $\frac{dx}{dt} = 0.5$:

$$\frac{dV}{dt} = 3(6)^2(0.5)$$

$$\frac{dV}{dt} = 54 \text{ cm}^3/\text{s}$$

Final Answer: $\frac{dV}{dt} = 54 \text{ cm}^3/\text{s}$

Important Note

Sign check: an increasing dimension has a positive rate, and a decreasing dimension has a negative rate.